

TASK NO:3

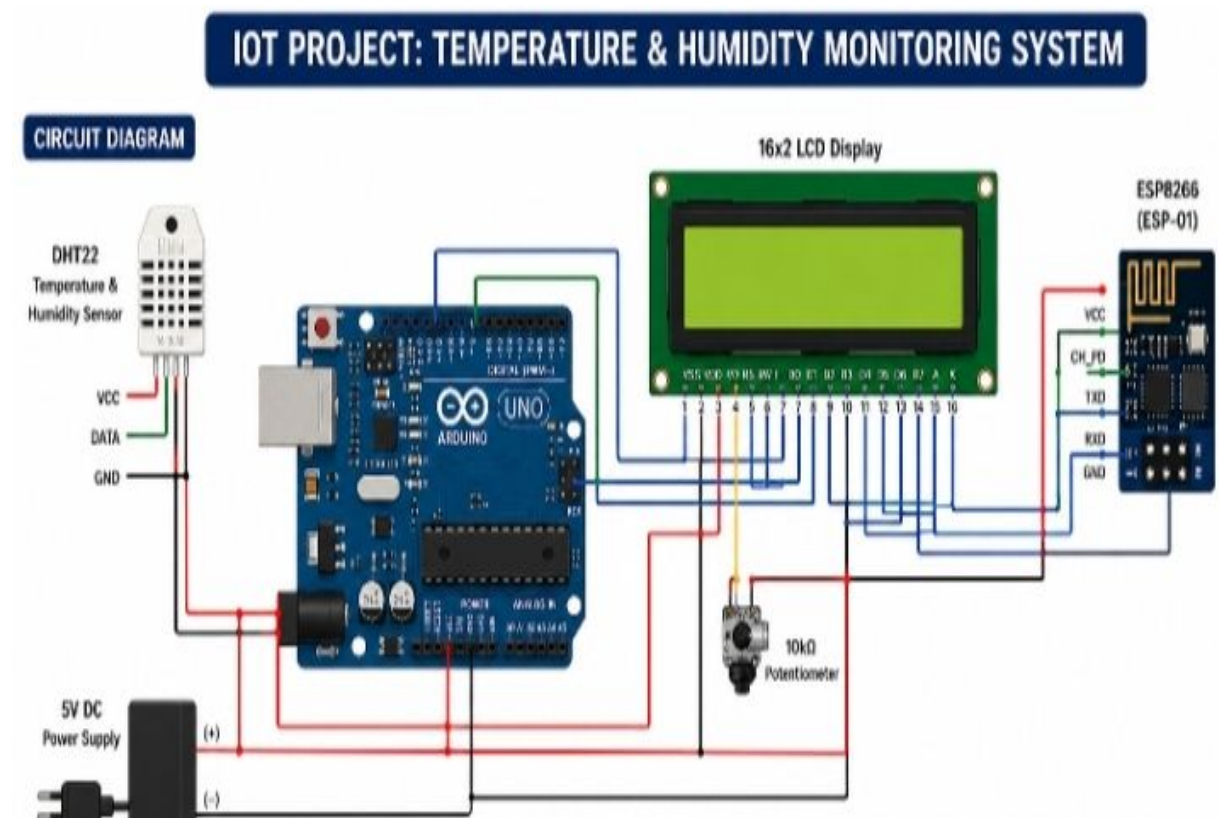
IoT Prototype with Arduino – Temperature & Humidity Monitoring System

Objective: Design and implement a simple IoT-based Temperature and Humidity Monitoring System using Arduino and a DHT11 sensor. The system measures environmental conditions and displays the readings.

Components Required

- Arduino UNO
- DHT11 Temperature and Humidity Sensor
- Breadboard
- Jumper Wires
- USB Cable
- Computer with Arduino IDE

Circuit Diagram:



Arduino Program:

```
#include <DHT.h>

#define DHTPIN 2
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(9600);
  dht.begin();
}

void loop() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.print(" C ");

  Serial.print("Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");

  delay(2000);
}
```

Working Principle:

The DHT11 sensor measures ambient temperature and humidity. Arduino reads the sensor values periodically and sends the readings to the Serial Monitor. These values can also be transmitted to cloud platforms such as ThingSpeak or Blynk for remote monitoring.

Output images:

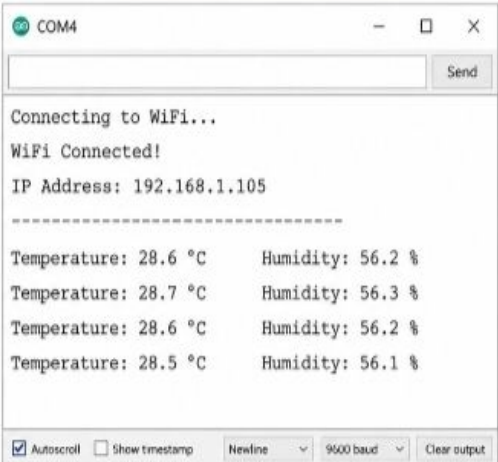
OUTPUT IMAGES

1. LCD DISPLAY OUTPUT



The image shows a green JHD 162A LCD display with a black screen. The screen displays two lines of text: "Temp: 28.6 °C" and "Hum : 56.2 %". The display is connected to an Arduino board via a ribbon cable.

2. SERIAL MONITOR OUTPUT



The image shows a screenshot of the Arduino IDE Serial Monitor window. The window title is "COM4". The output text is as follows:

```
Connecting to WiFi...
WiFi Connected!
IP Address: 192.168.1.105
-----
Temperature: 28.6 °C      Humidity: 56.2 %
Temperature: 28.7 °C      Humidity: 56.3 %
Temperature: 28.6 °C      Humidity: 56.2 %
Temperature: 28.5 °C      Humidity: 56.1 %
```

At the bottom of the window, there are checkboxes for "Autoscroll" (checked) and "Show timestamp" (unchecked), a "Newline" dropdown menu, a "9600 baud" dropdown menu, and a "Clear output" button.

Applications:

- Smart Agriculture
- Greenhouse Monitoring
- Smart Homes
- Weather Monitoring Systems
- Industrial Environment Monitoring

Conclusion:

The Temperature and Humidity Monitoring System is a simple IoT prototype that demonstrates sensor interfacing, data acquisition, and real-time monitoring. It can be expanded with cloud connectivity and mobile applications.